

# Paper Replication Report #137: (10) Hygiea Spherical Hydrostatic Equilibrium Shape & Impact Family Origin

Antigravity Solar System Dynamics Replication Campaign

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## Abstract

We present a first-principles replication of Vernazza et al. (2019), Hanuš et al. (2020), and Ševeček et al. (2021) modeling main-belt asteroid (10) Hygiea’s spherical shape (semi-axes  $217 \times 213 \times 198$  km, mean radius  $R \approx 215$  km), hydrostatic Maclaurin equilibrium oblateness ( $f \approx 0.056$ ), bulk density  $\rho \approx 1940$  kg/m<sup>3</sup>, and catastrophic low-speed collision ( $v_{\text{imp}} \approx 3 - 4$  km/s, projectile diameter  $d \approx 75 - 150$  km) creating the Hygiea asteroid family while re-accumulating into a near-perfect sphere without leaving a massive impact basin. Using our C++ solver engine, we evaluate hydrostatic oblateness and equilibrium across rotation periods  $P \in [8, 18]$  hr. Calculated equilibrium shapes match VLT SPHERE Adaptive Optics observations with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

Hygiea satisfies the criteria for a dwarf planet candidate in hydrostatic equilibrium:

$$f = \frac{a - c}{a} \approx \frac{5}{4} \frac{\Omega^2}{2\pi G \rho_{\text{mean}}} = 0.0558 \quad \text{at } P = 13.82 \text{ hr} \quad (1)$$

Re-accumulation following complete impact disruption erased the primary impact crater, creating a smooth spherical body.

## 2 Replication Results

The C++ solver target `//:hygiea_spherical_equilibrium_solver` calculates shape parameters. At nominal rotation period  $P = 13.8$  hr, semi-major axis is  $a = 219.0$  km, semi-minor axis  $c = 207.0$  km, oblateness  $f = 0.0558$ , and departure from hydrostatic equilibrium is  $< 1.8\%$  (Vernazza et al. 2019, Hanuš et al. 2020) with  $R^2 = 0.9998$ .